PACKAGE FORMULA: HEXAGONAL STRUCTURE AND PERIODIC TABLE OF BLACK HOLES IN MACRO SCALE

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Model: Package Formula / Slowik's Package Formula - 1 Package = 2.8 M☉

ABSTRACT:

This work presents an author's gravitational concept called "Package Formula", postulating the existence of discrete structural units of collapse. In model of main astrophysical nurseries, collapse of stellar matter to singularity-free states is postulated. As opposed to this paradigm, present model shows that gravitationally stable macro-objects (so-called black holes) form geometrically ordered, quantized multi-package structures, constituting a direct analogue of the periodic table of elements in macro scale. Division of collapse energy occurs in a precise balance between spin of geometric core and surrounding accretion disk.

1. THEORETICAL BASIS: QUANTUM OF GRAVITATIONAL MASS

Fundamental assumption of the Package Formula is existence of minimal, stable structural unit of gravitational collapse, amounting exactly to 2.8 M☉ (Solar masses). This value represents elementary gravitational "quantum" or "package", from which more complex cosmic macro-structures are built. All excessive mass of collapsing star, which does not constitute a multiple of this basic structural portion, becomes force of the medium emitted outward, supplying directly the surrounding accretion disk.

In process of macro-object formation, system energy is distributed according to strict gravitational-rotational balance:

$$M_{total} = n \times (2.8\ M_\odot) + M_{disk}$$

where n denotes an integer number specifying macroscopic atomic number of object, and M_disk is mass of accretion disk engaged in process of macro-synthesis of heavy elements.

2. DIMENSIONLESS SPIN PARAMETER AND GEOMETRY OF CORE

In real Universe every black hole exhibits rotation described by dimensionless spin parameter a* in range from 0 to 1. State of ideal non-rotating (a* = 0, Schwarzschild metric) constitutes only mathematical simplification. Real objects tend to physical limit of speed of light (a* → 1.0). Dynamic analysis shows that quantized mass packages arrange into symmetric geometric structures, generating powerful angular momentum.

Table 1: Classification and gravitational periodic table of macro-objects.

Astronomical Object	Observed Mass (M☉)	Number of Packages (n)	Geometric Configuration	Measured Spin (a*)	Chemical Analogue
Basic Unit	2.8	1	Gravitational Quantum	—	Hydrogen (^{1}H)
GRO J1655-40	ca. 6.3	2	Binary system + Disk	0.65 – 0.80	Helium (^{4}He)
GRS 1915+105	ca. 10 – 18	6	Hexagon (Stable Hex)	0.98 – 1.00	Carbon (^{12}C)

3. HEXAGONAL STRUCTURE (STABLE HEX) ON EXAMPLE OF GRS 1915+105

The most spectacular confirmation of Package Formula symmetry are parameters of object GRS 1915+105. At total mass estimated in upper range about 18 M☉, simple and elegant equations of core structure division are revealed:

$$18\ M_\odot / 3\ M_\odot \approx 6 \text{ and } 6 \times 2.8\ M_\odot = 16.8\ M_\odot$$

Mass difference remains ideally allocated in rotational energy and dynamic accretion disk. Number 6 represents unique spatial symmetry - hexagon. In fixed core of carbon ^{12}C consisting of 6 protons and 6 neutrons (represented often as stable arrangement of three alpha particles) exhibits analogically extremely durable configuration properties. GRS 1915+105 constitutes macroscopic cosmic carbon, spinning with extreme frequency 1150 times per second, which brings its spin close to limiting value a*

≈ 1.0.

4. CONCLUSIONS AND COSMIC SYNTHESIS OF MATTER

Application of hexagonal geometry and principle of mass quantization sheds new light on mechanisms of Milky Way evolution, analyzed in macro scale, consists of 74% hydrogen and 24% helium, constituting a reflection of elementary atomic composition. Annihilation and destructive, gigantic gravitation of spinning macro-nuclei (such as binary and six-package ones) acts as highest-performance cosmic accelerator. Friction and pressure prevailing in accretion disks generate conditions for continuous synthesis of heavy elements on macro scale, organizing Universe into coherent, fractal whole, where microworld of atoms meets its gravitational copy in macro scale.

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